Scheduling Concurrent Training 48 versus 72 h after Simulated Match Play: Effects on Neuromuscular Function and Fatigue

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ABSTRACT

CROSS, R., R. LOVELL, P. W. MARSHALL, D. NORRIS, and J. C. SIEGLER. Scheduling Concurrent Training 48 versus 72 h after Simulated Match Play: Effects on Neuromuscular Function and Fatigue. *Med. Sci. Sports Exerc.*, Vol. 55, No. 2, pp. 301–310, 2023. **Introduction:** Scheduling concurrent training (CT) during the in-season microcycle in field-based team sport is driven by prematch and postmatch recovery. This study examined the neuromuscular function, fatigue, and soreness responses to CT administered 48 h (match day (MD) + 2) versus 72 h (MD + 3) after match. **Methods:** Ten male recreational-level team sport athletes were monitored daily during two 5-d microcycles, which began with a simulated match (Soccer-specific Aerobic Field Test ($SAFT^{90}$)) and CT performed either 48 or 72 h after match. Maximal voluntary force, quadriceps maximum EMG, voluntary activation, muscle contractile function (evoked twitch responses), muscle soreness, and fatigue were assessed immediately before and after the $SAFT^{90}$, and every 24 up to 96 h after match. Outcome measures were also assessed immediately after CT. The CT consisted of an intermittent sprint protocol and a lower limb resistance training session separated by 1 h. **Results:** Immediately after the $SAFT^{90}$ in both conditions, maximal voluntary force was below baseline (mean change (Δ), −14.6% ± 10.0%; $P = 0.03$), recovering 48 h post. Quadriceps contractile function (Δ, −31.5% ± 11.4%; $P = 0.003$) and voluntary activation (Δ, −8.9 ± 6.2%; $P = 0.003$) were also hampered after the $SAFT^{90}$, recovering 24 h post in both conditions. In addition, the $SAFT^{90}$ elicited elevated levels of fatigue and muscle soreness that recovered 24 h after the $SAFT^{90}$ before increasing at 72 and 96 h post in the MD + 2 and MD + 3 conditions, respectively. **Conclusions:** Recovery of fatigue was only observed at the end of the microcycle when CT was prescribed on MD + 2. Therefore, CT scheduled early (MD + 2) in the microcycle might avoid compromising forthcoming match preparation. **Key Words:** SCHEDULING, CONCURRENT TRAINING, FATIGUE, QUADRICEPS CONTRACTILITY, VOLUNTARY ACTIVATION

Throughout the competitive phase of the season in field-based team sports, training is managed in weekly microcycles with the goal of maintaining the level of physical adaptation attained during the preseason. To achieve this, each microcycle is scheduled in a manner that considers fatigue, acute decrements in physical function, and recovery from both match play and various training stimuli. A 48-h postmatch recovery period is commonly prescribed with training often condensed to the middle of the microcycle, resulting in a high prevalence of concurrent training (CT) sessions, which involve both field-based training and resistance exercise performed on the same day (1). However, there is a dearth of empirical and mechanistic evidence regarding how such scheduling approaches alter neuromuscular function and fatigue across a microcycle.

Fatigue is a self-reported symptom commonly observed after physical and cognitive exertion (2). After team sport match play, persistent fatigue and decrements in the capacity of muscle to maximally and explosively produce force have been observed alongside impairments in contractile function and voluntary activation (VA), which take up to 48–72 h to fully resolve (3,4). This lengthy recovery process is suggested to be driven by myofibrillar damage and the associated inflammatory response that is elicited by the high eccentric demand of match play (3,4). Accordingly, practitioners avoid resistance, cardiorespiratory, and field-based exercise prescription during the 48-h postmatch recovery period (1), and tend to prescribe these modalities concurrently (i.e., within 1–2 h) in the middle of the microcycle.

Although CT is seemingly a necessary feature of the in-season microcycle, the fatiguability of muscle is an inevitable consequence. Studies suggest that complete recovery of muscle function and soreness takes up to 48 h (5–7). Given

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this recovery time course from CT (48 h), the scheduling of such sessions across the middle (match day (MD) + 3 to MD − 2) of the microcycle might impact subsequent match preparation (1). This has been supported by observations that delaying the prescription of resistance and eccentric-based injury prevention training (48–72 h after match) elevated creatine kinase (CK) and muscle soreness during the latter phases of the microcycle (8,9). These indirect measures of muscle damage (CK and soreness) have been linked with reduced external loads during field-based training (10–12). Furthermore, this midweek prescription has the potential to compromise match performance in short (3–5 d) microcycles; hence, important training stimuli (resistance training (RT) and injury prevention) are often sacrificed (13,14).

A potential solution to the challenges of scheduling training within the congested in-season microcycle might be reducing the number of days for recovery after match. Although scheduling training closer to match play might delay postmatch recovery, the available evidence suggests that low-volume and low-intensity eccentric (9) and resistance (high-velocity) (8) exercise performed within close proximity to match play (24 h post) does not delay or exacerbate postmatch recovery of muscle function. However, it is unknown if administering a training session of higher volume and intensity, such as team sports CT, 48 h after match extends the postmatch recovery time course and/or compromises the training stimulus (i.e., volume of cardiorespiratory and resistance exercise). Moreover, there is no available evidence examining changes in both neuromuscular function and fatigue in relation to different microcycle scheduling strategies. The aim of this study was to examine the acute impact of scheduling CT on different days of the microcycle upon measures of neuromuscular function (i.e., maximal force, rate of force development (RFD), maximum surface EMG (sEMG), VA, and contractility), fatigue, and muscle soreness.

METHODS

Participants

Ten male recreational team sport athletes (field hockey, n = 1; soccer, n = 7; Australian Rules football, n = 1; rugby league, n = 1) with a history of RT (age, 23 ± 4 yr; stature, 177.5 ± 5.0 cm; body mass, 78.5 ± 11.0 kg) volunteered to participate in this study after providing written informed consent. All participants were free from injury and/or illness and participated in at least one sports-specific training session and one competitive match per week. This study was approved by the Western Sydney University Human Research Ethics Board and was performed in accordance with the principles in the Declaration of Helsinki.

Experimental Design

Participants attended the laboratory on 11 separate days in the off-season phase of competition. In the initial session, participants were familiarized with all testing procedures including femoral nerve stimulation and maximal voluntary contractions where feedback on effort was provided, and each component of a team sports–specific CT session (intermittent sprint protocol (ISP) and lower limb RT session) and a simulated match (Soccer-specific Aerobic Field Test ($SAFT^{90}$)). Seven days later, participants returned to the laboratory for the first of two experimental microcycle conditions, where participants were monitored across 5 consecutive days. Each microcycle began with the participants completing a simulated match. Participants then performed a CT session either 2 d (MD + 2; 48 h post) or 3 d (MD + 3; 72 h post) after the simulated match (Fig. 1). Outcome measures were collected immediately before (BASELINE) and after ($POST_{SAFT}$) the simulated match, on all nonexercise days as well as before (PRE_{CT}) and after ($POST_{CT}$) the CT session. The order of the conditions was randomized and counterbalanced with the total volume of running performed

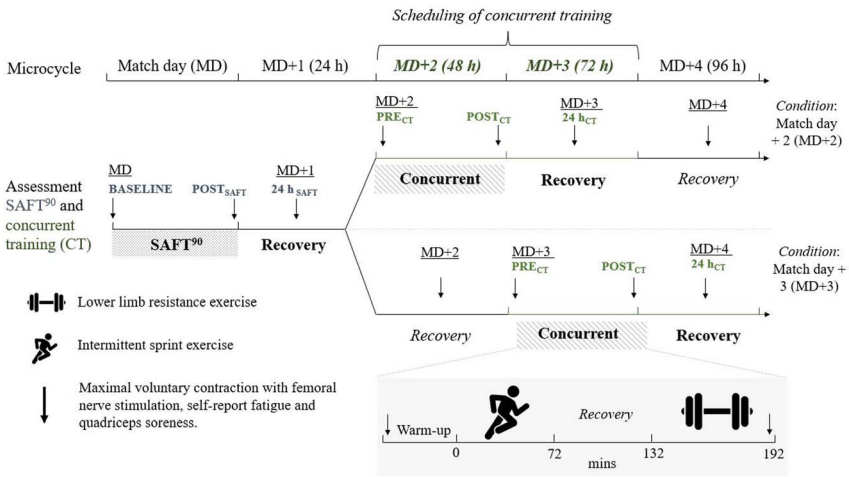

FIGURE 1—Schematic of study design. The microcycle examination looks at the changes in neuromuscular function and fatigue across the week by examining immediately pre- (BASELINE) and post-$SAFT^{90}$ as well as at 24-h increments for the entire 5-d microcycle (24, 48, 72, and 96 h). The acute response to the CT (PRE_{CT}, $POST_{CT}$, 24 h_{CT}) session was also examined. Plus symbol represents the training session in respect to the number of days after match.

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within the intermittent sprint component of CT matched between conditions.

In the 24 h before the match simulation, participants were asked to refrain from all forms of high-intensity exercise. Furthermore, participants were asked to abstain from any additional training during the 5-d microcycles. Normal nutritional patterns were recorded throughout the first microcycle, and participants were encouraged to use this record to replicate their dietary intake to the best of their abilities during the subsequent week.

Simulated Team Sport Match (SAFT^{90})

The SAFT^{90} was used in the present study to provide a match stimulus at the beginning of each microcycle. Throughout the SAFT^{90}, participants navigate their way around a 20-m agility course with an audio file providing verbal cues detailing the movement type and intensity. Over 90 min (2 × 45 min halves), participants cover a total distance of 11.1 km comprising various movement types and speeds replicating match demands (15). Throughout the SAFT^{90}, a total of 18 maximal sprint efforts were performed, with the 10-m sprint time with a rolling 3-m start monitored throughout (SpeedLight; Swift Performance Equipment, Lismore, Australia). The external validity of the SAFT^{90} has been inferred from physiological responses (heart rate, muscle temperature) (16) and decrements in hamstring muscle function (17), sprint, and jump performance (16) comparable to those observed during actual match play (18,19).

Concurrent Training

Within the CT session, an ISP was performed 1 h before a lower limb RT session. The order of intermittent sprint and resistance exercise components of CT, as well as the 1-h rest period, was chosen to reflect common practice within the applied team sport setting (1,20).

Resistance training. A total of seven lower limb exercises were performed within the RT session. These included the barbell back squat (4 sets at 4–6 repetition maximum loading), barbell lunge (8 reps, 4 sets at 50% body weight (BW)), single-leg Romanian deadlift (8 reps, 4 sets at 12%–15% BW), forward hurdle jumps (5 reps, 4 sets; BW), sideways hurdle jumps (5 reps, 4 sets; BW), squat jumps (5 reps, 4 sets; BW), and Nordic hamstring curls (5 reps, 4 sets; BW). This session was designed through consultation with industry experts and relevant scientific literature to reflect a common in-season RT prescription (21,22). The repetition maximum loading for the back squat was determined in the familiarization session, and the interset rest period for all exercises was 1.5–2 min.

Intermittent sprint protocol. The ISP, chosen to reflect the high-intensity intermittent nature of field-based training in team sport, was adapted from previous literature (23). The ISP was composed of two halves, each containing two identical 15-min blocks of running separated by a 1-min passive rest period, whereas the two halves were separated by 10 min (Fig. 2). These blocks of running were performed over a 20-m indoor running track and included one 15-m sprint with a 5-m deceleration every minute for 15 min. Between each maximal sprinting effort, participants performed continuous laps of jogging, running, or walking for the remainder of the minute. After each sprint, only one type of submaximal exercise was performed and rotated in the aforementioned order, ceasing approximately 10 s before the upcoming sprint. On the 6th and 12th minutes of each 15-min block, the submaximal running was substituted with eight consecutive double-foot horizontal bounds where participants were encouraged to cover as much distance as possible (24,25). The total number of laps covered throughout the entire 60-min period was monitored to calculate the total distance covered, and the 20-m track was marked in 1-m increments to record the distance covered in the eight consecutive bounds. Total distance covered in the first experimental condition completed by each participant was then matched during the subsequent visit to standardize the ISP stimulus, thus allowing for more control during MD + 2 and MD + 3. Fifteen-meter sprint time was monitored throughout the full ISP (SpeedLight, Swift Performance Equipment).

Neuromuscular Assessment

Maximal voluntary isometric contractions. Maximal voluntary isometric contractions (MVCs) of the knee extensors with superimposed femoral nerve stimulation were

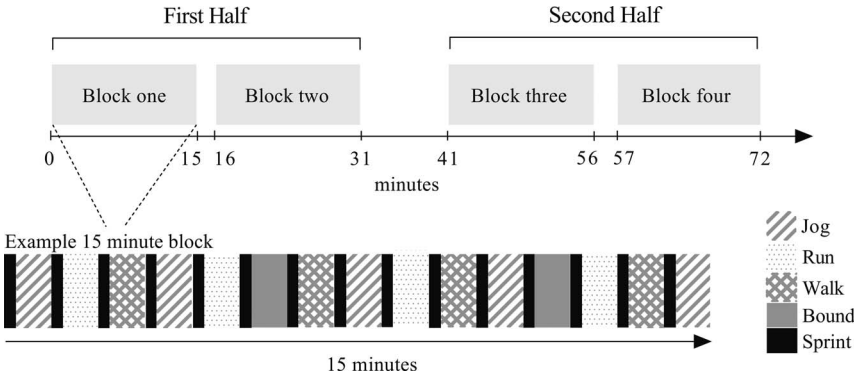

FIGURE 2—Schematic representation of the ISP.

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performed before and immediately after the $SAFT^{90}$ as well as 24, 48, 72, and 96 h post. In addition, MVCs were performed immediately after the CT session (Fig. 1). All MVCs were performed on a custom-built adjustable chair with a strain gauge force transducer (PT Global, Sydney, Australia) where participants were seated with their hip and knee joints at 90°. The force transducer was attached to a noncompliant cuff that was firmly secured to participants' right leg, 2 to 3 cm superior to the lateral malleolus. All MVCs (except for $POST_{SAFT}$ and $POST_{CT}$) were preceded by four submaximal isometric knee extensor contractions at increasing intensities (2 × 50%, 2 × 75% perceived effort). Three MVCs, with 1-min rest between efforts, were performed at each measurement time point with participants instructed to perform each MVC as "fast and forcefully" as possible and maintain the contraction for 3 to 4 s. Standardized verbal encouragement was provided throughout all MVCs. Force signals were continuously sampled from the strain gauge at 1000 Hz (Powerlab; ADI Instruments, Sydney, Australia; 10-Hz low-pass filter), and a predetermined calibration factor was applied to convert the voltage signal to force (N).

Surface EMG. During all MVCs, electrical activity of the vastus lateralis (VL) and vastus medialis (VM) was recorded using surface electrodes (Ag/AgCI surface electrodes; Maxsensor; Medimax Global, Sydney, Australia). To prepare the skin for electrode placement, excess hair was removed, and the area was abraded with a fine sandpaper and cleaned with an isopropyl alcohol swab. Two electrodes were placed on the VM, 3 to 4 cm superior to the medial border of the patella (interelectrode distance, 10 mm) arranged in the presumed direction of the muscle fibers. To record muscle activity of the VL, the distal electrode was placed at the midpoint between the greater trochanter and the lateral aspect of the patella on the muscle belly, and the second electrode was placed 6–10 cm above the distal electrode, on the proximal portion of the VL (26). A ground electrode was placed on the fibular head, and all electrode placements were marked with indelible ink for consistency throughout all experimental sessions. sEMG signals were sampled at 4000 Hz (ADI Instruments) using the ML138 Octal BioAmp (common-mode rejection ratio >85 dB at 50 Hz, input impedance of 200 MΩ) with 16-bit analog-to-digital conversion. A fourth-order Bessel filter between 20 and 500 Hz was applied to the recorded signals, which were smoothed with a 100-ms root mean square calculation.

Femoral nerve stimulation. To locate the femoral nerve for cathode placement, a rubber insulated probe was moved around the femoral triangle using a 30-mA stimulation intensity. The location that produced the largest VL and VM M-wave and muscle twitch was marked with a felt-tip pen, and an electrode (Ag/AgCI; 2-cm diameter) was applied. An anode was constructed of aluminum foil (5 × 9 cm) covered with conducting gel (Ten20; Weaver and Company, Aurora, CO) and placed on the lateral aspect of the hip midway between the greater trochanter and the iliac crest. Ten-hertz supramaximal doublets (2-ms square wave pulse) were applied to the femoral nerve during all MVCs using a high-voltage (400 V) constant current stimulator (Digitimer DS7AH; Digitmer, Hertfordshire, United Kingdom). The stimulation intensity applied was determined before each testing day by incrementally increasing the intensity until a plateau in the twitch force and VL and VM M-waves was obtained. The average maximal currents across the weekly microcycle were 95.0 ± 17.8 mA (BASELINE), 93.3 ± 12.3 mA (MD + 1), 97.5 ± 21.4 mA (MD + 2), 99.2 ± 27.8 mA (MD + 3), and 90.0 ± 17.3 mA (MD + 4). Supramaximal stimulation was achieved by increasing the intensity by 20%. During all MVCs when force reached a visible plateau, superimposed doublets were delivered to the femoral nerve with a resting potentiated twitch ($Q_{.pot.tw.doublet}$) evoked 2 to 3 s after each MVC using 10-Hz doublets.

Perceived Fatigue and Quadriceps Soreness

To measure fatigue perception, the five-item fatigue subsection of the Profile of Mood States (POMS-fatigue) was used. The sum of all scores in five categories (worn out, fatigued, exhausted, sluggish, weary; 0 = not at all, 1 = a little, 2 = moderately, 3 = quite a bit, 4 = extremely) determined the fatigue value (27). Using a visual analog scale (0 = no soreness; 10 = worst soreness), quadriceps soreness was examined both during seated rest (PASSIVE) and during an unweighted leg extension (ACTIVE) with no pressure applied to the muscle. Sessional ratings of perceived exertion (sRPE) were determined after each exercise session using the 0 to 10 category ratio RPE scale (28), which was then multiplied by the duration of the session (29).

Data Processing

From the force recordings of the MVCs, the following were analyzed: maximal voluntary force (MVF; excluding the point of stimulation), and maximal RFD, calculated as the greatest average 10-ms slope during the first 200 ms from contraction onset. The average of the three MVCs was used for analysis.

VA was estimated using the following equation:

$$\text{VA (\%)} = 100 - \left(\text{Superimposed Twitch} \times \left(F_{\text{pre-stim}}/\text{MVF}\right)/Q_{.\text{pot.tw.doublet}}\right) \times 100$$

where the superimposed twitch is the difference between the voluntary force immediately before the twitch ($F_{\text{pre-stim}}$) and the maximal force recorded during the twitch, MVF is maximal voluntary force during the MVC (excluding the twitch), and $Q_{.\text{pot.tw.doublet}}$ is the peak force of the resting potentiated twitch evoked with a 10-Hz doublet (30).

The following variables were analyzed from the resting potentiated twitch: 1) the maximal force, 2) time-to-peak twitch (TPT) and half-relaxation time (1/2RT), and 3) maximal RFD, calculated as the greatest average 10-ms slope. Twitch onset for the calculation of both RFD and TPT was systematically identified as the point at which force reached 2.5 N above zero.

The VL and VM sEMG collected during all MVCs was analyzed for the peak-to-peak amplitude of the first M-wave of

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the 10-Hz doublet (raw sEMG signal) and the maximal amplitude of the root mean square signal (excluding the point of stimulation) in the 250 ms before the evoked twitch. Maximal sEMG amplitude was normalized to the first M-wave of the 10-Hz doublet (EMG/M; %).

Reliability

Mean within-subject coefficient of variations (CV) from the three baseline MVCs were 5.6% ± 5.4% for MVF, 21.8% ± 16.7% for maximal voluntary RFD (vRFD), 5.5% ± 3.1% for $Q_{.\text{pot.tw.doublet}}$ force, 7.6% ± 6.3% for $Q_{.\text{pot.tw.doublet}}$ RFD, 9.9% ± 9.8% for 1/2RT, 5.9% ± 10.2% for TPT, 14.8% ± 12.6% for VL sEMG, 14.0% ± 11.3% for VM sEMG, and 2.6% ± 2.3% for VA. Mean between-day (baseline MVCs from both conditions), within-subject CV values were 7.2% ± 3.6% for MVF, 30.5% ± 16.3% for maximal vRFD, 4.0% ± 3.2% for $Q_{.\text{pot.tw.doublet}}$ force, 10.4% ± 6.0% for $Q_{.\text{pot.tw.doublet}}$ RFD, 15.9% ± 12.1% for 1/2RT, 7.4% ± 8.6% for TPT, 17.3% ± 10.0% for VL sEMG, 11.1% ± 8.2% for VM sEMG, and 1.9% ± 1.4% for VA.

Statistical Analysis

Repeated-measures ANOVA was used to examine changes in each outcome measure across the full microcycle (BASELINE; $\text{POST}_{\text{SAFT}}$; 24, 48, 72, and 96 h) and between conditions (MD + 2 and MD + 3). Within this analysis, the 48- and 72-h time points reflect the pre-CT (PRE_{CT}) assessment for the MD + 2 and MD + 3 conditions, respectively. A separate parallel analysis was performed to compare the acute response to the CT session (PRE_{CT}, POST_{CT}, 24 h_{CT}) between conditions (MD + 2 and MD + 3). All analyses were performed using IBM SPSS v25 (IBM Corp, Armonk, NY). Sphericity was assessed using the Mauchly test, and if violated, the Greenhouse–Geisser correction was used to adjust the degrees of freedom. *Post hoc* tests were completed with Bonferroni's correction if main significant effects of time, condition, or the interaction between time and condition were observed. Significance level was set at $P \leq 0.05$. All data are presented as mean ± SD. Effect sizes (ES) were calculated (Cohen's *d*) for pairwise comparisons. The magnitude of the ES was deemed small (≥0.2), medium (≥0.6), and large (≥0.8) (31). A sensitivity analysis was performed to identify the smallest ES we could observe with 80% power assuming an α of 0.05, ρ (correlation of between repeated measures) of 0.5, and our sample size of 10. This analysis was performed using the Superpower package in R studio, specifying a within-subject design. Results from this analysis revealed sensitivity to detect Cohen's *D* ES differences of 0.71 (moderate/large) for condition–time effects (32).

RESULTS

Match Simulation and CT Characteristics

The average heart rate response, sprint time, and session RPE for the SAFT^{90} and CT (ISP and RT) sessions are presented in Table 1. The total distance covered during the ISP in both conditions was 6834 ± 665 m, and no significant difference between conditions was observed for the total distance covered in the eight consecutive bounds throughout the entire ISP (MD + 2: 131 ± 22 m; MD + 3: 130 ± 19 m; $P = 0.41$).

TABLE 1. Characteristics of the simulated match (SAFT^{90}) and the individual components of the CT session (ISP and RT) for both conditions (MD + 2 and MD + 3).

	Average Heart Rate (bpm)	sRPE (AU)	Sprint Time (s)		Total Volume and Load
			10 m	15 m	
SAFT^{90}					
MD + 2	156 ± 8	735 ± 123	1.73 ± 0.08		
MD + 3	156 ± 12	700 ± 126	1.77 ± 0.11		
Contrast	$P = 0.89$	$P = 0.46$	$P = 0.08$		
ISP					
MD + 2	153 ± 13	518 ± 121		2.77 ± 0.16	
MD + 3	154 ± 16	490 ± 139		2.76 ± 0.16	
Contrast	$P = 0.84$	$P = 0.30$		$P = 0.44$	
RT					
MD + 2	123 ± 10	348 ± 104			3654 ± 425
MD + 3	125 ± 14	336 ± 104			3633 ± 410
Contrast	$P = 0.49$	$P = 0.56$			$P = 0.34$

Data are mean ± SD ($n = 10$). Total volume and load were calculated by multiplying the load lifted by the number of reps and sets for all exercises (load × reps × sets).

Full Microcycle

MVF and vRFD. There was no significant time–condition ($P = 0.27$) effect observed for MVF. However, a main effect of time ($P = 0.003$) revealed that MVF was reduced from baseline immediately after the SAFT^{90} (mean change (Δ), −14.6% ± 10.0%; $P = 0.03$; ES, 0.68; Fig. 3) and remained below baseline at 24 h post (Δ, −7.4% ± 5.6%; $P = 0.008$; ES, 0.30), before recovering at 48 h. There was no condition effect ($P = 0.18$) for MVF. No significant time–condition ($P = 0.60$), time ($P = 0.32$), or condition ($P = 0.59$) effects were observed for vRFD.

Muscle contractile function. There was no time–condition effect for $Q_{.\text{pot.tw.doublet}}$ force ($P = 0.19$). A significant main effect of time ($P < 0.001$) revealed that $Q_{.\text{pot.tw.doublet}}$ amplitude declined by −31.5% ± 11.4% ($P = 0.003$; ES, 1.32; Fig. 3) below baseline immediately after the SAFT^{90}, recovering at 24 h post. There was no effect of condition for $Q_{.\text{pot.tw.doublet}}$ force ($P = 0.70$). Following a similar postmatch trend, no time–condition ($P = 0.33$) effect was observed for $Q_{.\text{pot.tw.doublet}}$ RFD. However, a main effect of time ($P < 0.001$) revealed that $Q_{.\text{pot.tw.doublet}}$ RFD decreased by −18.7% ± 14.4% ($P = 0.05$; ES, 0.83) immediately after the SAFT^{90}, recovering 24 h post. There was no condition effect for $Q_{.\text{pot.tw.doublet}}$ RFD ($P = 0.63$). For TPT, there was no significant time–condition ($P = 0.24$), time ($P = 0.34$), or condition ($P = 0.76$) effect. No significant time–condition effect was observed for 1/2RT ($P = 0.39$). However, a main effect of time ($P = 0.001$) revealed a −23.4% ± 13.8% ($P = 0.001$; ES, 1.57) postmatch reduction, which recovered at 24 h post. No condition ($P = 0.96$) effect was observed for 1/2RT.

VA and sEMG. No time–condition ($P = 0.75$) effects were observed for VA. However, a significant main effect of time ($P < 0.001$) revealed that VA declined 8.9% ± 6.2% ($P = 0.003$; ES, 1.43) below baseline immediately after simulated match play, recovering 24 h post. There was no significant condition ($P = 0.27$) effect for VA.

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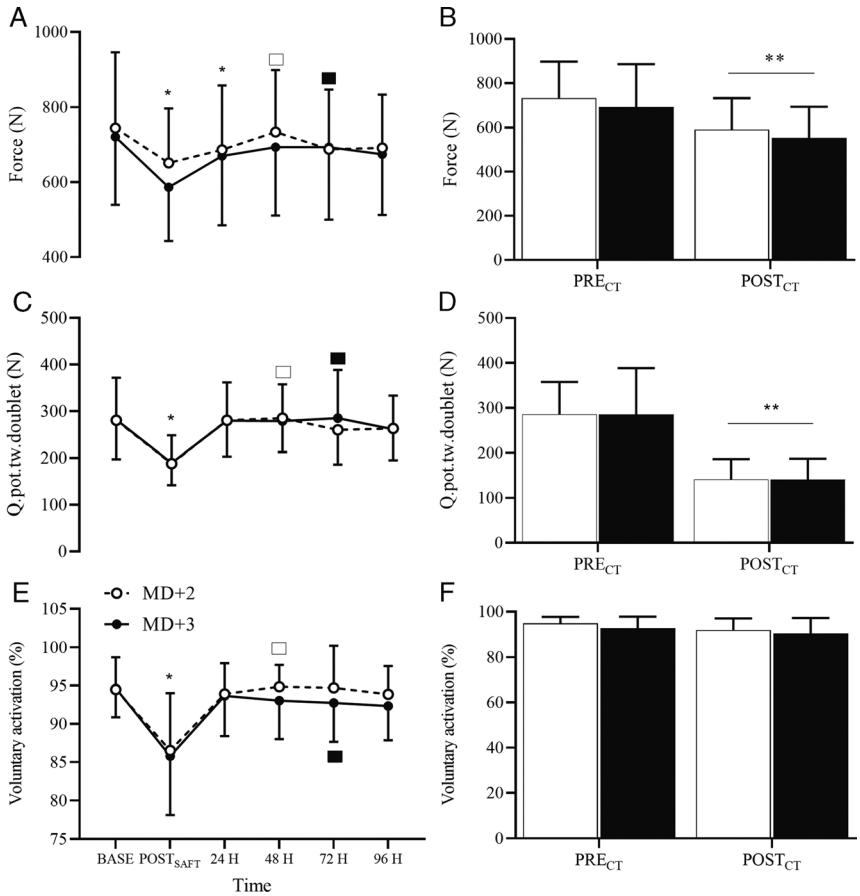

FIGURE 3—MVF (A), quadriceps potentiated twitch force (C), and VA (E) across the full weekly microcycle and immediately before and after CT (B, MVF; D, $Q_{.pot.tw.doublet}$; F, VA). Data are mean ± SD (n = 10). * denotes significant difference from BASELINE; P < 0.05. □ corresponds to PRE$_{CT}$ in the MD + 2 condition, and ■ corresponds to PRE$_{CT}$ in the MD + 3 condition. Each time point contains data from both conditions.

A significant time–condition (P = 0.017) effect was observed for VL sEMG. *Post hoc* comparisons reveal that at 72 h after the SAFT^{90}, VL sEMG was higher in the MD + 2 (8.74% ± 3.34% Mmax) condition compared with MD + 3 (7.53% ± 3.28% Mmax; P = 0.028; ES, 0.37). There was no significant main effect of time (P = 0.63) or condition (P = 0.71) for VL sEMG.

There was no significant interaction effect (P = 0.42) observed for VM sEMG. However, a significant main effect of time (P = 0.014) was observed for VM sEMG. *Post hoc* analysis revealed that VM sEMG was lower than baseline at 96 h after simulated match play (BASELINE, 9.97% ± 1.91% Mmax; 96 h, 8.69% ± 2.15% Mmax; P = 0.041; ES, 0.63). There was no condition effect for VM sEMG (P = 0.68).

No time–condition effect (P = 0.63) was observed for VM M-wave amplitude. However, a significant main effect of time (P < 0.001) revealed that VM M-wave amplitude declined (Δ, 23.07% ± 19.3%; P = 0.007; ES, 1.06) below baseline immediately after simulated match play, recovering 24 h post. There was no condition (P = 0.28) effect for VM M-wave amplitude. For VL M-wave amplitude, no time–condition (P = 0.22), time (P = 0.28), or condition (P = 0.69) effects were observed.

Fatigue and quadriceps soreness. A significant time–condition effect (P < 0.001) was observed for POMS-fatigue, indicating that at 48 and 72 h after match, fatigue was higher in the MD + 2 condition (48 h, 5.6 ± 2.8 AU; 72 h, 7.5 ± 2.9 AU) compared with the MD + 3 condition (48 h, 3.3 ± 2.5 AU (P = 0.05; ES, 0.87); 72 h, 2.0 ± 2.5 AU (P = 0.001; ES, 2.01); Fig. 4). In addition, at 96 h post match, perceived fatigue was higher in the MD + 3 condition (7.9 ± 2.0 AU; MD + 2, 4.7 ± 4.9 AU; P = 0.05; ES, 0.86), returning to baseline in the MD + 2 (P = 1.0; ES, 0.17) condition but not in the MD + 3 condition (P = 0.024; ES, 1.4). A significant main effect of time (P < 0.001) revealed a post-SAFT^{90} increase in fatigue (BASELINE, 3.7 ± 4.0 AU; POST$_{SAFT}$, 15.1 ± 3.0 AU; P < 0.001; ES, 3.18), which returned to baseline at 24 h post. There was no significant main effect of condition for fatigue (P = 0.47).

A significant interaction effect was observed for both active (P < 0.001) and passive (P < 0.001) muscle soreness. *Post hoc* analysis revealed higher active and passive muscle soreness at 72 h after match in the MD + 2 (active, 4.3 ± 2.7 cm; passive, 3.3 ± 2.7 cm) condition compared with MD + 3 (active, 1.5 ± 1.8 cm (P = 0.001; ES, 1.22); passive, 1.0 ± 1.3 cm (P = 0.003; ES, 1.09); Fig. 4). A significant main effect of time was observed for both active (P = 0.001) and passive (P < 0.001) quadriceps soreness, with *post hoc* comparisons revealing elevated muscle soreness above baseline immediately after simulated match play (active: Δ, 3.3 ± 2.2 cm (P = 0.019; ES, 1.50); passive: Δ, 2.8 ± 2.0 cm

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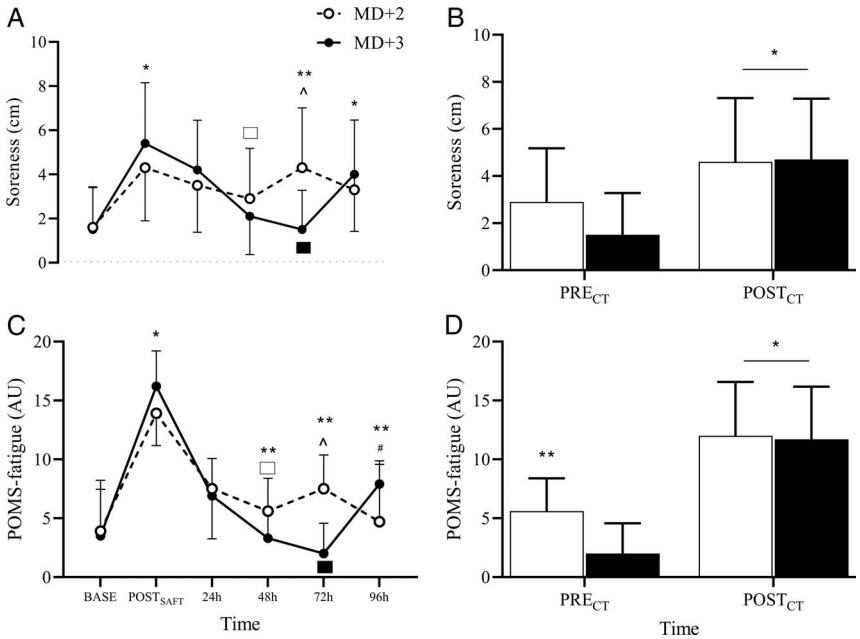

FIGURE 4—Active quadriceps muscle soreness (A) and fatigue as determined from the total score of the POMS (C) across the entire microcycle and immediately before and after CT (B, active soreness; D, POMS-fatigue). Data are mean ± SD (n = 10). * denotes significant difference from BASELINE in both conditions, $P < 0.05$. ^ denotes significant difference from BASELINE in the MD + 2 condition only; # denotes significant difference from BASELINE in MD + 3 condition only, $P = 0.024$; ** denotes significant difference between conditions, $P < 0.05$. □ corresponds to PRE$_{CT}$ in the MD + 2 condition, and ■ corresponds to PRE$_{CT}$ in the MD + 3 condition. Each time point contains data from both conditions.

(P = 0.029; ES, 1.32)). Active and passive soreness returned to baseline 24 h post match, before displaying a secondary increase above baseline at 72 h in the MD + 2 condition (active: Δ, 2.7 ± 1.8 cm (P = 0.001; ES, 1.18); passive: Δ, 2.0 ± 2.2 cm (P = 0.003; ES, 0.89)) and at 96 h post-SAFT[90] in both conditions (active: Δ, 2.1 ± 2.5 cm (P = 0.018; ES, 0.97); passive: Δ, 1.6 ± 1.8 cm (P = 0.027; ES, 0.85)).

CT Session

Neuromuscular response. Before CT (PRE$_{CT}$) in both conditions, all changes in neuromuscular function had returned to baseline. Immediately after CT, there were significant decrements in MVF (Δ, −20.8% ± 12.9%; P = 0.009; ES, 0.97; Fig. 3), $Q_{.pot.tw.doublet}$ maximal force (Δ, −47.1% ± 17.8%; P = 0.003; ES, 2.07; Fig. 3), $Q_{.pot.tw.doublet}$ RFD (Δ, −41.5% ± 17.3%; P < 0.001; ES, 1.81), and VM M-wave amplitude (Δ, −16.6% ± 13.5%; P < 0.001; ES, 0.88). There was no significant difference between conditions, and all neuromuscular variables returned to baseline at 24 h post-CT (Fig. S1, Supplemental Digital Content, Acute response to the CT session for MVF, quadriceps potentiated twitch force, and VA, http://links.lww.com/MSS/C717).

Fatigue and quadriceps soreness. At the commencement of CT, POMS-fatigue was higher in the MD + 2 (5.6 ± 2.8 AU) condition compared with MD + 3 (2.0 ± 2.6 AU; P = 0.006; ES, 1.33). Immediately after CT, fatigue was elevated above baseline (MD + 2: Δ, 8.1 ± 5.8 AU (ES, 1.82; P = 0.01); MD + 3: Δ, 8.2 ± 5.3 AU (ES, 1.94; P = 0.005)) and remained elevated at 24 h post-CT (Δ, 4.0 ± 3.8 AU; ES, 1.21; P = 0.04), with no significant difference between conditions at both the immediately post-CT and 24 h post-CT time points. Active and passive soreness was elevated above baseline immediately after CT (active: Δ, 3.1 ± 2.1 cm (P = 0.008; ES, 1.38); passive: Δ, 2.9 ± 2.1 cm (P = 0.012; ES, 1.43)) and remained above baseline at 24 h post-CT (active: Δ, 2.6 ± 2.4 cm (P = 0.001; ES, 1.31); passive: Δ, 1.8 ± 2.1 cm (P = 0.006; ES, 0.97)), with no significant difference between conditions (Fig. S2, Supplemental Digital Content, Acute response to the CT session for active quadriceps muscle soreness and fatigue as determined from the total score of the POMS, http://links.lww.com/MSS/C717).

DISCUSSION

This study examined changes in neuromuscular function, fatigue, and muscle soreness in response to CT performed either 48 or 72 h after simulated match play during a 5-d microcycle. MVF, VA, and muscle contractility were all reduced immediately after simulated match play, returning to baseline values before the CT session in both conditions, before displaying a decline immediately after CT. Simulated match play elicited fatigue, which resulted in higher self-reported fatigue at the commencement of CT performed on MD + 2 compared with the training session performed on MD + 3. This difference in fatigue did not impact the performance of the CT session, nor did it exacerbate the fatigue, heart rate, or sRPE response. Recovery of neuromuscular function from CT was observed within 24 h regardless of proximity to match play,

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whereas fatigue remained elevated from baseline at 24 h post-CT. Fatigue remained above baseline at the end of the microcycle (96 h after simulated match play) when CT was performed on MD + 3.

Simulated match play elicited a reduction in MVF of the quadriceps with concomitant reductions in VM M-wave amplitude, VA, and quadriceps twitch characteristics. The magnitude of reduction observed in knee extensor MVF output (14%) and VA (8%) is consistent with findings from both actual and simulated match play (MVC, 11%–20%; VA, 7%–15%) (3,4,33–35). In contrast, the 31% reduction in twitch amplitude observed immediately after match is larger than previously reported after both simulated (14%–23%) (4,33,34) and actual match play (9%–14%) (3,35). This discrepancy might be linked to a number of methodological differences such as the stimulation methodology used, the duration of time between the end of match play, and the assessment of neuromuscular function (immediately after compared with ≥40 min) (3,35), and the specific characteristics of the simulations and match play (4,33). Regardless, all measures had recovered within 24–48 h after the SAFT^{90}, which is a shorter recovery time when compared with actual match play (48–72 h) (3,35). The absence of sports-specific actions (kicking, tackling, and jumping) in the SAFT^{90}, previously observed to influence the neuromuscular response to match play (36), might have been responsible for the rapid recovery. However, previous studies examining changes in neuromuscular function after actual match play have demonstrated that a substantial level of restoration occurs by 24 h post, with twitch amplitudes and VA remaining below baseline by only 6% (~13 N decrement) and 4.7%, respectively (3). The functional significance of the small deficits previously reported at 24 h after match is unclear, particularly when considering the reported measurement error associated with these techniques (CV, 2.2% VA, 5.3% $Q_{\text{pot.tw}}$) (4,33).

Complete recovery of neuromuscular function was observed before the commencement of CT in both conditions. Immediately after CT, decrements in MVF and muscle contractility were observed, with no change in VA and sEMG, irrespective of CT scheduling. These findings support previous CT literature and suggest that impairments in the capacity of muscle to maximally produce force after CT are mediated by factors occurring at or below the level of the neuromuscular junction (5). For example, alterations in sarcolemma excitability as evidenced by the reduction in peak-to-peak amplitude of the VM M-wave might, in part, have contributed to changes in muscle contractility. In addition, although not measured within the present study, changes in contractility might be associated with tissue damage as demonstrated by increased levels of CK previously reported after the same CT session administered in the present study (5). All changes in neuromuscular function after CT recovered at 24 h, regardless of scheduling. These results suggest that the neuromuscular recovery kinetics after CT are unaffected by the provision of only 48-h recovery after match. However, we must acknowledge our small sample size and lack of a control condition, and thus, we cannot rule out the possibility of an effect going undetected. In addition, further research is required to examine the chronic implications of scheduling CT 48 h after simulated match play.

In contrast to the trends in neuromuscular function, weekly changes in self-reported fatigue were influenced by the scheduling of CT. Interestingly, at 48 h after simulated match play, self-reported fatigue was higher in the MD + 2 condition. This trend was not driven by any differences in the acute fatigue response from match play or any differences in the level of activity performed within the 48-h recovery period. Given that neuromuscular function had fully recovered, the discrepancies in the level of perceived fatigue 48 h after match might be linked to situational and psychological factors (37). It has been reported that the sensation of fatigue might be elicited by expectations of upcoming task demands and by the memory of previous experiences relevant to the ensuing activity (37). In the present study, pretraining fatigue was lower when CT was performed 72 h after match compared with 48 h. However, consistent with previous literature examining the fatigue response to consecutive days of training (38), elevated pretraining fatigue in the presence of full neuromuscular recovery (MD + 2 condition) did not lead to an exacerbated reporting of fatigue immediately after CT.

Although the fatigue response immediately post-CT was not influenced by scheduling, administering CT 72 h after simulated match play resulted in fatigue and elevated muscle soreness at 96 h after match, which, in a 5-d microcycle, represents the day before the upcoming match. This finding supports previous literature examining the scheduling of injury prevention programs, where the delayed prescription (72 h after match) resulted in elevated muscle soreness in the latter phase of the microcycle (MD + 5) (9). Case study examples of microcycle structure have revealed that skill and tactic-based training sessions are common in the latter stages of the microcycle (14), which might be negatively impacted by the persistent elevation of fatigue and muscle soreness (10,12). The increased perception of fatigue at the commencement of match play or training has been reported to negatively impact external training load variables (e.g., high-intensity running, maximal running velocity, and PlayerLoad), perhaps leading athletes to self-regulate the intensity of exercise and manage the perceived degree of effort (10,12). Therefore, prescribing training earlier in the microcycle (48 h after match), as observed in rugby union (39), might assist in ensuring that important match preparation, training stimuli, and match play occurring in the latter stages of the microcycle are not negatively influenced or even omitted (13,14) because of elevated fatigue and muscle soreness. However, despite an absence of evidence to suggest that this early microcycle training prescription negatively impacts performance and the acute neuromuscular response, athletes might be less willing to undertake such training sessions earlier in the week given their elevated perceptions of fatigue and muscle soreness. Further research is warranted to confirm suggestions that elevated

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fatigue in the latter stages of the microcycle might impact upcoming match and training performance.

Limitations

The match simulation used in the present study allowed for control over match running variables; however, the simulation fails to account for the added physical demands of sports-specific movements such as kicking, tackling, and jumping. The absence of such sports-specific actions likely contributed to the rapid postsimulation recovery of neuromuscular function, despite some of the immediate postsimulation changes in neuromuscular function being comparable to competitive match play. Similar limitations are also present in the use of a simulated field-based training session (ISP) despite the total distance and sRPE being comparable to that observed in field-based training sessions during the in-season phase of competition (40,41). Furthermore, the translation of the findings in this study are limited to noncontact sports, as the match and field-based training simulations used did not simulate contact, which is known to delay postmatch recovery kinetics (42).

The generalizability of the results of this study to an elite athlete context might be questioned because of the level of athlete examined and number of participants. Participants were recreational-level athletes from mixed field-based team sports backgrounds (field hockey, soccer, Australian Football, rugby league). Notwithstanding, baseline quadriceps MVC (733 N) of the participants within this study and the decrement in force (−14%) after the match simulation are comparable to that previously reported in semiprofessional athletes (baseline MVC, 632–726 N; post-match reduction in force, 9%–15%) (3,4,35). We also acknowledge that characteristics of CT (i.e., order of sessions, between-session rest period, volume, and intensity) likely vary among team sports; thus, the generalizability of the results of this study is specific to the CT prescription administered. Moreover, we acknowledge the relatively small sample size, which was a result of the extensive participant time commitment (e.g., 11 laboratory visits) and limitations in funding. As such, only findings with moderate to large effects (ES, >0.71) were considered when interpreting the results of this study.

Finally, the use of a 10-Hz doublet to estimate VA and contractile function is also a limitation of this study. Because of the 100-ms duration between interpolated twitches evoked during a maximal voluntary contraction, the two stimuli of a 10-Hz doublet may not summate as compared with a 100-Hz doublet. As a result, the measured interpolated twitch may be similar to that evoked by a single stimulus. However, previous research has demonstrated that VA and quadriceps twitch amplitudes determined using both 10- and 100-Hz stimulation techniques elicit similar amplitudes and responses after an exercise stimulus (43).

CONCLUSIONS

Scheduling CT 48 h after simulated match play did not impact performance or the acute neuromuscular, fatigue, and soreness response. Alternatively, the scheduling of CT 72 h after match resulted in elevated levels of fatigue and muscle soreness in the latter stages of the microcycle despite full recovery of neuromuscular function. This persistent fatigue has the potential to impact preparation (field-based skill and tactic training sessions) for the upcoming match. The results of this study suggest that administering CT earlier in the weekly microcycle would allow more time for the prescription of training stimuli that might otherwise be omitted because of microcycle congestion.

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